

Unit III — Graphs and Isomorphism

Bipartite Knowledge Graphs and Permutation-Similarity Certificates

GROUP 4 — Discrete Mathematics

Mini R&D Report — August 28, 2026

This is an educational mathematics assignment demonstrating the application of discrete mathematics to a hypothetical medical diagnosis system. It does not constitute medical advice.

1. Title of the Case Study

Modelling Symptom-Disease Knowledge as a Bipartite Graph and Proving Isomorphism Between Two Representations

Overview

Medical knowledge bases represent relationships between symptoms and diseases as networks. This report models such a knowledge base as a bipartite graph $G = (S \cup D, E)$, where S is a set of symptom vertices, D is a set of disease vertices, and edges represent 'is indicative of' relationships. We then prove that two visually distinct drawings of this graph are in fact the same knowledge base — that is, they are isomorphic.

The proof uses graph invariants (vertex count, edge count, degree sequence, bipartiteness) and an explicit edge-preserving bijection, verified by the algebraic certificate $A_2 = P \cdot A_1 \cdot P^T$ involving biadjacency and permutation matrices.

2. Problem Statement

Two medical editors maintain the same symptom-disease knowledge base in different tools. Editor 1 uses a bipartite layout (symptoms left, diseases right). Editor 2 uses a circular layout (nodes arranged in a ring). When the files are merged, the reviewer sees two different drawings and must determine:

1. Is the underlying graph the same? (Are G_1 and G_2 isomorphic?)
2. What graph invariants can we check to build confidence?
3. Can we exhibit an explicit edge-preserving bijection $f: V(G_1) \rightarrow V(G_2)$?

4. Can we produce an algebraic certificate (permutation matrix P) such that $A_2 = P \cdot A_1 \cdot P^T$?
5. What is the practical significance for medical knowledge base management?

The two representations are:

Representation	Layout	Vertex Order
G_1 (Bipartite)	Symptoms left, diseases right	$(S_1, S_2, S_3, S_4 \mid D_1, D_2, D_3)$
G_2 (Circular)	Nodes in a ring	$(S_4, D_1, S_1, D_3, S_2, D_2, S_3)$

Table 1: The two representations to be compared.

3. Real-World Context

The Knowledge Base Merging Problem

In hospitals and research institutions, symptom-disease knowledge bases are maintained by multiple editors using different tools:

- Editor 1 might use a graph database (Neo4j) with a bipartite visualisation
- Editor 2 might use a spreadsheet exported to a circular network diagram
- Editor 3 might use a clinical decision support system with yet another layout

When these knowledge bases need to be merged — for example, when two hospitals consolidate their triage systems — the reviewer must determine whether the underlying knowledge is identical despite different visual representations.

Why Isomorphism Matters

- De-duplication: Prevents counting the same clinical rule twice after a merger
- Conflict detection: If the graphs are NOT isomorphic, the editors have introduced different knowledge — this is a real conflict that needs resolution
- Version control: Tracking changes to a knowledge base requires identifying which edges were added, removed, or modified — this requires comparing graph structures, not layouts
- Regulatory compliance: The EU AI Act requires documenting the knowledge base; two drawings of the same graph should produce one documentation entry, not two

The Bipartite Structure

Medical knowledge naturally forms bipartite graphs because symptoms and diseases are distinct entity types. A symptom never 'indicates' another symptom directly — the relationship always crosses from the symptom set to the disease set. This bipartite structure is a powerful constraint that simplifies isomorphism checking.

4. Mathematical Concept Used

Graph Theory Foundations

Concept	Definition	Relevance
Graph $G = (V, E)$	Set of vertices V and edges E	The knowledge base structure
Bipartite graph	$V = S \cup D, E \subseteq S \times D$	Symptoms don't connect to symptoms
Degree $\deg(v)$	Number of edges incident to v	Invariant for isomorphism
Degree sequence	Sorted list of all degrees	Necessary condition for isomorphism
Handshake lemma	$\sum \deg(v) = 2 E $	Verification check
Isomorphism	Bijection f preserving adjacency	Core question of this report
Biadjacency matrix	Matrix A where $A[i,j] = 1$ iff $(s_i, d_j) \in E$	Algebraic representation
Permutation matrix	Matrix P with exactly one 1 per row/column	Certificate of reordering

Table 2: Graph theory concepts used in this report.

Isomorphism: Formal Definition

Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic ($G_1 \cong G_2$) if there exists a bijection $f: V_1 \rightarrow V_2$ such that for all $u, v \in V_1$:

$$\{u, v\} \in E_1 \iff \{f(u), f(v)\} \in E_2$$

In words: f preserves adjacency. If two vertices are connected in G_1 , their images under f are connected in G_2 , and vice versa.

Graph Isomorphism Problem (GI)

Determining whether two graphs are isomorphic is a famous problem in computational complexity:

- GI is in NP (a candidate bijection can be verified in polynomial time)
- GI is NOT known to be NP-complete (one of the few natural problems in this state)
- GI is NOT known to be in P (no polynomial-time algorithm is known for general graphs)
- For bipartite graphs of fixed size (like ours with $|V|=7$), exhaustive checking is trivial

Biadjacency Matrices

For a bipartite graph with symptom set $S = \{s_1, \dots, s_m\}$ and disease set $D = \{d_1, \dots, d_n\}$, the biadjacency matrix A is an $m \times n$ matrix where:

$A[i, j] = 1$ if $(s_i, d_j) \in E$, and 0 otherwise

Key property: if G_2 is obtained from G_1 by reordering symptoms and diseases, then $A_2 = P_S \cdot A_1 \cdot P_D^T$, where P_S and P_D are permutation matrices for the symptom and disease orderings respectively.

5. Mathematical Formulation

Graph Definition

$G = (V, E)$ where:

Set	Elements	Count
V (vertices)	$S \cup D$	$ V = 7$
S (symptoms)	$\{S_1 \text{ Fever, } S_2 \text{ Cough, } S_3 \text{ Rash, } S_4 \text{ Fatigue}\}$	$ S = 4$
D (diseases)	$\{D_1 \text{ Influenza, } D_2 \text{ Measles, } D_3 \text{ Anemia}\}$	$ D = 3$
E (edges)	$\{(S_1, D_1), (S_1, D_2), (S_2, D_1), (S_3, D_2), (S_4, D_1), (S_4, D_3)\}$	$ E = 6$

Table 3: Formal definition of the knowledge graph G .

Vertex Degrees

Vertex	Type	Incident Edges	Degree
S_1 (Fever)	Symptom	S_1-D_1, S_1-D_2	2
S_2 (Cough)	Symptom	S_2-D_1	1
S_3 (Rash)	Symptom	S_3-D_2	1
S_4 (Fatigue)	Symptom	S_4-D_1, S_4-D_3	2
D_1 (Influenza)	Disease	$S_1-D_1, S_2-D_1, S_4-D_1$	3
D_2 (Measles)	Disease	S_1-D_2, S_3-D_2	2
D_3 (Anemia)	Disease	S_4-D_3	1

Table 4: Degree sequence for all vertices.

Degree sequence (sorted): (3, 2, 2, 2, 1, 1, 1)

Handshake lemma check: $\sum \deg(v) = 2+1+1+2+3+2+1 = 12 = 2 \times 6 = 2|E|$ ✓

The Bijection f

The natural bijection $f: V(G_1) \rightarrow V(G_2)$ maps each vertex to its prime-labelled counterpart:

G_1 Vertex	G_2 Vertex (f)	Type
S_1	S_1'	Symptom $\rightarrow$ Symptom
S_2	S_2'	Symptom $\rightarrow$ Symptom
S_3	S_3'	Symptom $\rightarrow$ Symptom
S_4	S_4'	Symptom $\rightarrow$ Symptom
D_1	D_1'	Disease $\rightarrow$ Disease
D_2	D_2'	Disease $\rightarrow$ Disease
D_3	D_3'	Disease $\rightarrow$ Disease

Table 5: The edge-preserving bijection f .

Verification: f maps symptoms to symptoms and diseases to diseases. Since G_1 and G_2 have the same edge list (just reordered), every edge $\{u,v\}$ in G_1 maps to $\{f(u),f(v)\}$ in G_2 . ✓

6. Analysis / Solution

Step 1: Invariant Comparison

Before attempting to find an isomorphism, we check necessary conditions (graph invariants):

Invariant	G_1 (Bipartite)	G_2 (Circular)	Match?
Vertices $ V $	7	7	✓
Edges $ E $	6	6	✓
Degree sequence	(3,2,2,2,1,1,1)	(3,2,2,2,1,1,1)	✓
Bipartite?	Yes (S D)	Yes (S D)	✓
Max degree	3 (D_1)	3 (D_1')	✓
Min degree	1 (S_2, S_3, D_3)	1 (S_2', S_3', D_3')	✓

Table 6: Invariant comparison between G_1 and G_2 .

All invariants match. This is necessary but not sufficient for isomorphism. (Non-isomorphic graphs can share the same degree sequence.) We now exhibit an explicit bijection.

Step 2: Edge-Preserving Bijection Verification

We verify that f preserves all 6 edges:

Edge in G_1	Image under f	Edge in G_2 ?
$\{S_1, D_1\}$	$\{S_1', D_1'\}$	✓
$\{S_1, D_2\}$	$\{S_1', D_2'\}$	✓
$\{S_2, D_1\}$	$\{S_2', D_1'\}$	✓
$\{S_3, D_2\}$	$\{S_3', D_2'\}$	✓
$\{S_4, D_1\}$	$\{S_4', D_1'\}$	✓
$\{S_4, D_3\}$	$\{S_4', D_3'\}$	✓

Table 7: Edge preservation under f . All 6 edges map correctly.

Since f is a bijection on 7 vertices and preserves all 6 edges (and therefore all $21 - 6 = 15$ non-edges), f is an isomorphism. $G_1 \cong G_2$. ■

Step 3: Algebraic Certificate — Permutation Matrices

The circular representation G_2 reorders symptoms as (S_4, S_3, S_2, S_1) and diseases as (D_3, D_2, D_1) . This corresponds to reversal permutation matrices:

Symptom permutation $r_S = [3, 2, 1, 0]$ (maps index i to $3-i$):

$P_S = [[0,0,0,1], [0,0,1,0], [0,1,0,0], [1,0,0,0]]$

Disease permutation $r_D = [2, 1, 0]$ (maps index j to $2-j$):

$P_D = [[0,0,1], [0,1,0], [1,0,0]]$

Biadjacency matrix A_1 (order $S_1..S_4 \mid D_1..D_3$):

	D_1	D_2	D_3
S_1	1	1	0
S_2	1	0	0
S_3	0	1	0
S_4	1	0	1

Table 8: Biadjacency matrix A_1 for G_1 .

Biadjacency matrix A_2 (order $S_4..S_1 \mid D_3..D_1$):

	D_3	D_2	D_1
S_4	1	0	1
S_3	0	1	0
S_2	0	0	1
S_1	0	1	1

Table 9: Biadjacency matrix A_2 for G_2 .

Verification: $A_2 = P_S \cdot A_1 \cdot P_D^T$

This equation is the algebraic certificate of isomorphism. It proves that G_2 is obtained from G_1 by simultaneously reordering symptoms (via P_S) and diseases (via P_D).

7. Diagram / Graph

Edge List with Clinical Semantics

#	Edge	Clinical Meaning	Direction
1	$S_1 - D_1$	Fever indicates Influenza	Symptom $\rightarrow$ Disease
2	$S_1 - D_2$	Fever indicates Measles	Symptom $\rightarrow$ Disease
3	$S_2 - D_1$	Cough indicates Influenza	Symptom $\rightarrow$ Disease
4	$S_3 - D_2$	Rash indicates Measles	Symptom $\rightarrow$ Disease
5	$S_4 - D_1$	Fatigue indicates Influenza	Symptom $\rightarrow$ Disease
6	$S_4 - D_3$	Fatigue indicates Anemia	Symptom $\rightarrow$ Disease

Table 10: Complete edge list with clinical interpretation.

Observations from the Graph Structure

- D_1 (Influenza) has degree 3 — it's connected to the most symptoms (Fever, Cough, Fatigue). This makes clinical sense: influenza presents with multiple common symptoms.
- S_1 (Fever) and S_4 (Fatigue) each have degree 2 — they're 'bridge' symptoms that indicate multiple diseases. This makes them less specific but more sensitive.
- D_3 (Anemia) has degree 1 — only Fatigue indicates it. This makes Fatigue the sole 'gateway' symptom for Anemia detection.
- S_2 (Cough) and S_3 (Rash) each have degree 1 — they're highly specific (each indicates exactly one disease), making them strong diagnostic indicators when present.
- The graph is connected: every vertex can reach every other vertex via a path. This means the knowledge base covers a coherent clinical space.

8. Interpretation of Results

What Isomorphism Means for the Knowledge Base

The isomorphism $G_1 \cong G_2$ proves that the two visual representations encode the same clinical knowledge. The only difference is the spatial arrangement of vertices — the connectivity structure is identical.

Layout vs. Structure

This result highlights a fundamental principle in graph theory: the visual layout of a graph carries NO structural information. Two drawings that look completely different can encode the same knowledge. This is why:

- Graph databases store adjacency lists, not coordinates
- Layout algorithms (force-directed, circular, hierarchical) are purely aesthetic
- Isomorphism testing must focus on connectivity, not appearance
- Version control for graphs must track edge changes, not layout changes

Clinical Implications

For the medical triage system:

- Two editors can work independently and merge their knowledge bases losslessly
- The isomorphism check can be automated: compare invariants first (cheap), then search for a bijection if invariants match
- Non-isomorphic graphs immediately flag real knowledge conflicts
- The bipartite structure simplifies the search: only cross-edges matter

9. Practical Application

The Isomorphism Explorer

The graph and its isomorphism are deployed in the live Isomorphism Explorer interactive card. Features include:

- Two side-by-side SVG renderings: bipartite layout (G_1) and circular layout (G_2)
- Toggle button to animate the transition between layouts using spring physics
- Click any vertex to highlight all incident edges and display its degree
- Invariant comparison panel showing $|V|$, $|E|$, degree sequence, and bipartiteness
- Isomorphic badge (✓) confirming the graphs are the same knowledge base
- All SVG coordinates computed from the graph model — never hardcoded

The 2-Hop Diagnosis Path

The bipartite structure enables a powerful diagnostic pattern: 2-hop paths from patient to disease:

Patient → Symptom → Disease

If a patient has symptoms $\{S_1 \text{ Fever}, S_4 \text{ Fatigue}\}$, the system can follow edges:

- $S_1 \rightarrow D_1$ (Influenza), $S_1 \rightarrow D_2$ (Measles)
- $S_4 \rightarrow D_1$ (Influenza), $S_4 \rightarrow D_3$ (Anemia)
- Intersection: D_1 (Influenza) appears in both paths — highest confidence diagnosis

This path-based reasoning is a direct consequence of the bipartite structure and the isomorphism: it works identically in both representations.

10. Innovation / Proposed Improvement

Current Limitations

- Small graph: $|V| = 7$ is trivial for isomorphism testing. Real knowledge bases have hundreds of vertices.
- Static knowledge: the current graph doesn't change over time. Real systems need version control.
- Binary edges: an edge either exists or doesn't. Real clinical relationships have strengths (sensitivity, specificity).
- 2-partite only: real diagnosis involves patients as a third entity type.

Proposed Extensions

1. Graph Database Integration: Store the knowledge base in Neo4j or Amazon Neptune. Isomorphism checking becomes a merge tool for concurrent edits, using canonical labelling (nauty/Bliss algorithms) for efficient comparison.
2. 3-Partite Extension: Add a patient vertex set P , creating a 3-partite network $P \rightarrow S \rightarrow D$. Diagnosis becomes a 2-hop path problem: patient $\rightarrow$ symptom $\rightarrow$ disease. This enables personalised diagnosis based on individual symptom profiles.
3. Weighted Edges: Replace binary edges with clinical weights: $w(s, d)$ = sensitivity (probability of symptom s given disease d). The graph becomes a weighted bipartite graph, enabling probabilistic diagnosis.
4. Temporal Graphs: Add time stamps to edges: (s, d, t) meaning 'symptom s indicates disease d with onset time t '. This enables temporal reasoning: 'fever followed by rash within 48 hours suggests measles'.
5. Automated Isomorphism Detection: Implement a continuous integration check that compares knowledge base versions using the Weisfeiler-Lehman test (efficient for most practical graphs) with nauty as a fallback for counterexamples.

11. Conclusion

This report has modelled a symptom-disease knowledge base as a bipartite graph $G = (S \cup D, E)$ with $|V| = 7$, $|E| = 6$, and proved that two visually distinct representations are isomorphic.

Key findings:

- Graph invariants ($|V|$, $|E|$, degree sequence, bipartiteness) all match between G_1 and G_2
- An explicit edge-preserving bijection $f: V(G_1) \rightarrow V(G_2)$ was exhibited and verified on all 6 edges
- The algebraic certificate $A_2 = P_S \cdot A_1 \cdot P_D^T$ confirms the isomorphism via permutation similarity
- The bipartite structure is preserved: symptoms map to symptoms, diseases to diseases
- The result has practical significance for knowledge base merging, de-duplication, and version control

The isomorphism proof demonstrates that graph structure — not visual layout — is the invariant that matters for knowledge representation. Two drawings that look completely different can encode identical clinical knowledge. This principle underpins the design of graph databases, version control systems, and merge tools for medical knowledge bases.

Future work extends this to larger graphs (automated canonical labelling), weighted edges (probabilistic diagnosis), and 3-partite networks (patient-symptom-disease reasoning).

12. References

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All mathematical computations, proofs, and counts in this report are computationally verified and can be interactively tested at <https://ai-medical-diagnosis-site.vercel.app>